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Supplemental Information

Epistasis in a Fitness Landscape Defined by Antibody-Antigen Binding Free Energy

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Supplemental information for: Epistasis in a fitness landscape defined by antibody-antigen binding free energy

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Supplemental tables

domain	# of corresponding, catastraphic single mutants	epistasis type	# candidate mutants	# of mutants with sign epistasis (obs/exp)	# of mutants with reciprocal sign epistasis (obs/exp)	# of viable mutants with sign epistasis (obs/exp)
^{A}1H	$0, 1$	beneficial	825	22/0.51	10/0.08	22/0.15
^{B}1H	2	beneficial	172	13/0.10	13/0.02	11/0.03
^{C}1H	$0-2$	beneficial	997	35/0.61	23/0.10	33/0.18
^{D}1H	$0-2$	deleterious	53	3/0.61	0/0.10	0/0.18

Table S1: **Summary of CDR1H sign epistasis with double mutants, related to results section "Epistasis affects affinity".** Obs/exp denotes observed sign epistasis events versus expected epistasis events in a Gaussian noise model. Expected numbers below 0.01 are rounded up to 0.01. We measured 1058 CDR1H double mutants in total. Summaries are for sign epistasis that is A) beneficial and have at least 1 viable single mutation, B) beneficial but have no viable single mutations, C) beneficial, or D) deleterious. Mutants are only counted as sign or reciprocal sign epistatic if they pass statistical criteria, which may cause fewer reciprocal sign epistatic mutants to be counted than viable mutants with doubly catastrophic mutations. Double mutants are considered viable if their $K_d < 10^{-6}$ M.

domain	# of catastrophic mutations	epistasis type	# candidate mutants	# of mutants with sign epistasis (obs/exp)	# of mutants with reciprocal sign epistasis (obs/exp)	# of viable mutants with sign epistasis (obs/exp)
^{A}3H	$0, 1$	beneficial	698	5/0.41	3/0.07	3/0.05
^{B}3H	2	beneficial	359	3/0.21	3/0.03	3/0.02
^{C}3H	$0-2$	beneficial	1057	8/0.61	6/0.10	6/0.07
^{D}3H	$0-2$	deleterious	97	6/0.61	0/0.10	3/0.07

Table S2: **Summary of CDR3H sign epistasis with double mutants, related to results section "Epistasis affects affinity".** Obs/exp denotes observed sign epistasis events versus expected epistasis events in a Gaussian noise model. Expected numbers below 0.01 are rounded up to 0.01. We measured 1066 CDR3H double mutants in total. Summaries are for sign epistasis that is A) beneficial and have at least 1 viable single mutation, B) beneficial but have no viable single mutations, C) beneficial, or D) deleterious. Double mutants are considered viable if their $K_d < 10^{-6}$ M.

Supplemental figures

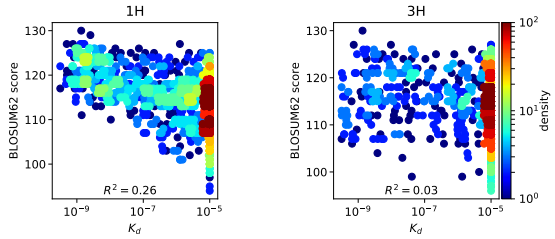

Figure S1: **BLOSUM62 predictions perform worse than PWM predictions, related to Figure 1.** While BLOSUM62 scores denoting distance to the wildtype sequence were predictive of K_d as previously observed (Łuksza et al., 2017), they performed much worse than PWM predictions (see Fig. 1).

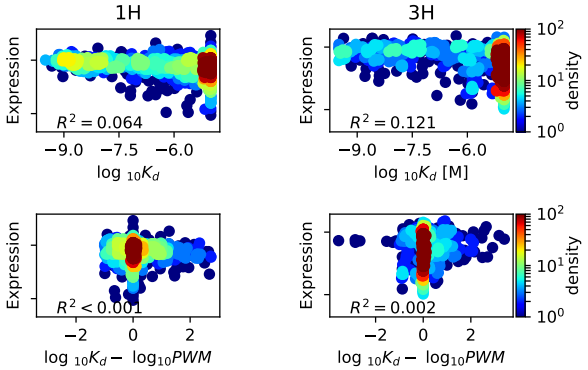

Figure S2: K_d **measurements are dissociated from expression measurements, related to Figure 1.** Expression is minimally correlated to K_d for CDR1H and CDR3H domains. The residual of $\log_{10}K_d$ further removes any relationship to expression. In comparison, the PWM based on K_D measurements explores a far higher fraction of observed variance (see Fig. 1C,D).

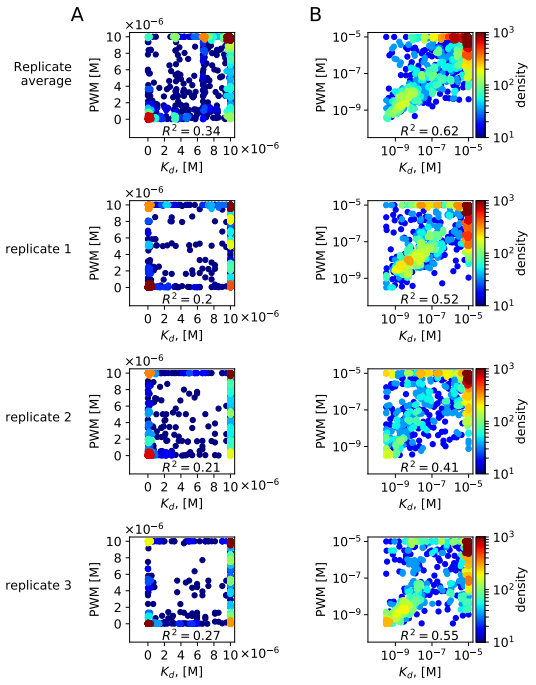

Figure S3: **Binding energy PWMs outperform K_d based models, related to Figure 1.** Comparison between data and model prediction of the binding affinity of multiple mutants using PWMs constructed from A) K_d and B) $F = \ln(K_d/c_0)$. Comparisons done on the average measured K_d obtained from 3 replicates (first row), and on each individual replicate (rows 2-4) show that averaged F measurements yield much better agreement than from any particular replicate. Although both predictions are very significant ($p < 10^{-5}$, F-test), the PWM based on F is much better, justifying its use in Fig. 1C,D.

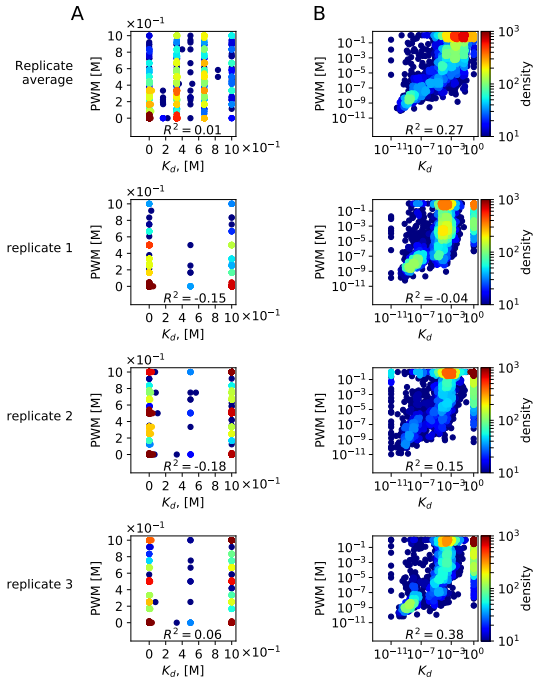

Figure S4: **Ignoring previous measurement boundaries yields poor PWM models, related to Figure 1.** Comparison between expanded data ($10^{-11} - 10^{0}$ M) and model prediction of the binding affinity of multiple mutants using PWMs constructed from A) K_d and B) $F = \ln(K_d/c_0)$. Comparisons were done on the average measured K_D obtained from 3 replicates (first row), and on each individual replicate (rows 2-4). R^2 denotes the coefficient determination (fraction of explained variance). R^2 denotes the coefficient determination (fraction of explained variance). Both predictions perform much worse ($R^2 = 0.01$ for K_d, $R^2 = 0.27$ for ln K_d) than when compared to models with the boundaries mirroring experimental precision, from $10^{-9.5} - 10^{-5}$ M as in Fig. S3, justifying the use of boundaries in Fig. 1C,D.

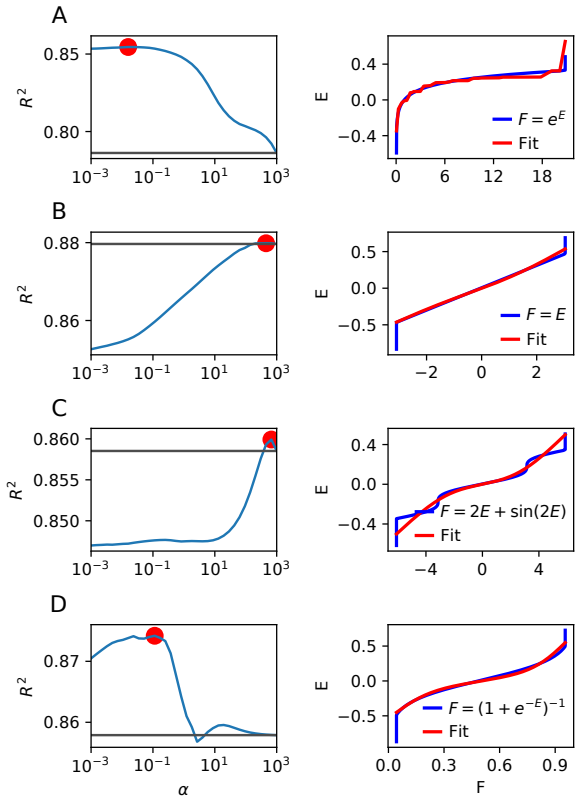

Figure S5: **Test of the inference method for learning the nonlinear scale on synthetic data, related to method section "Optimal nonlinear transformation of the free energy".** In order to test whether F, the binding energy, was nearly ideal for creating the additive model used in Fig. 1C,D, we developed an algorithm to find the ideal transformation of energy, and tested it on simulated data. A PWM was generated randomly, with terms drawn from a normal distribution, and applied to the sequences from our dataset to simulate a "true" score (E_{PWM}). Gaussian noise with 50% of the standard deviation of E_{PWM} was added. The score was the converted into a free energy $F = f(E)$ (Methods), where f, the inverse function of E, was a (**A**) linear, (**B**) exponential, (**C**) high frequency, or (**D**) logistic transformation. The range of measured F was cut off at the boundaries as in the data, resulting in the vertical lines in the right panels. Left panels: cross-validated fraction of explained variance, as a function of the regularization parameter α penalizing the second derivative of the function E. Large α implies very smooth functions, while small α allows for high-frequency variations. These results show the original PWM can be recovered as well as the non-linearity, except for its high-frequency components (C).

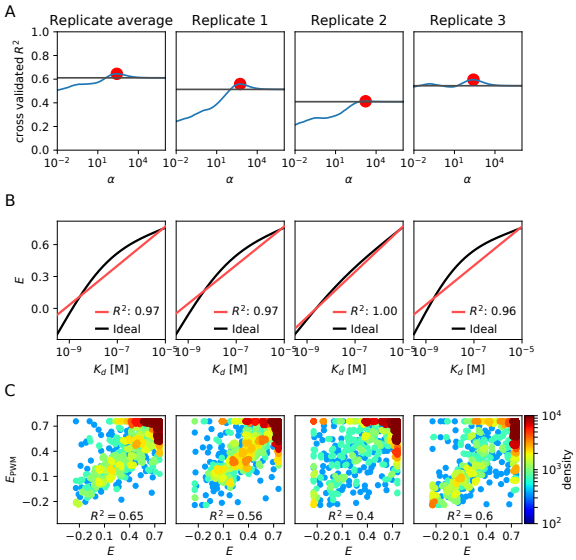

Figure S6: **Optimizing the non-linear scale E for the PWM model on real data, related to Figure 1. (A)** Cross-validated fraction of explained variance, as a function of the regularization parameter α penalizing the second derivative of the function E. (**B**) Optimized non-linear scale E as a function of $F = \ln(K_d/c_0)$ (black), compared to identity (red). As with the overall fit, the optimized PWM terms have 97% R^2 to the linear PWM terms. (**C**) Comparison between data and the PWM model with the optimal nonlinear scale. Comparisons were done on the average measured K_D obtained from 3 replicates (first column), and on each individual replicate (columns 2-4), and generally show little variation. These results show minimal improvements to the PWM used in Fig. 1C,D.

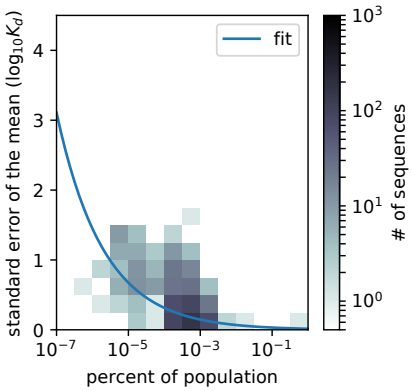

Figure S7: **Measurement error decreased with sequence abundance, related to Figure 2.** The percentage of the population that each clone represented was strongly anti-correlated with standard deviation of $\log_{10}K_d$ measurements. Excluding sequences with standard deviation = 0, we fit the model $\sigma = e^{\beta_0+\beta_1 \cdot p}$, where σ is the sample standard deviation of the $\log_{10}$ K_d, and p is the fraction of the population with a sequence. Our fit model predicts that the standard deviation decreases $10^{-0.33}$ fold for every decade increase in p. We used Z_{error} (see Fig. 2B) to distinguish the measurement error shown here, and epistatic variation.

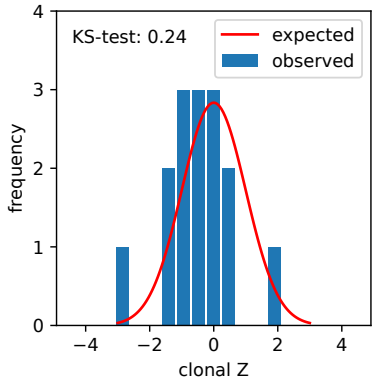

Figure S8: **Comparison of clonal Z-scores to estimate measurement error, related to Figure 2.** 19 antibody clones' K_d were previously measured by measuring their titration curves, 15 with K_d within boundaries of $10^{-9.5} - 10^{-5}$ M (Adams et al., 2016). Calculating Z-scores as $Z(\mathbf{s}) = \frac{F_{\text{tite}}(\mathbf{s}) - F_{\text{clonal}}(\mathbf{s})}{\sqrt{\sigma^2_{\text{tite}}(\mathbf{s}) + \sigma^2_{\text{clonal}}(\mathbf{s})}}$, we found that error between Tite-seq and clonal measurements did not deviate strongly from an expected normal distribution with mean = 0, standard deviation = 1 ($p = 0.24$, Kolmogorov-Smirnov test of normality), as we also saw with Z_{error} in Fig. 2B.

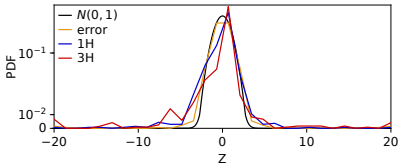

Figure S9: **Distribution of Z-scores including K_d within wide boundaries, related to Figure 2.** Expanding boundary measurements over a wide range ($10^{-11} - 10^{0}$ M) and including boundaries as in Fig. S4, the measurement error increases to standard deviation = 1.32. The fraction of variance explained due to noise increases substantially (to 26%), the fraction explained by PWM becomes quite small (to 32%), and the fraction explained by epistasis decreases slightly (to 42%). The average Z-score remained beneficial (=0.46). Correspondingly, the Z score standard deviation increased for CDR1H (2.26) and decreased for CDR3H (2.02). Restricting measurements to reliable boundaries yields more conservative estimates of epistasis wrt to the PWM, with measurement error that more closely follows the expected normal distribution with mean=0, standard deviation = 1, justifying this treatment in Fig. 2B.

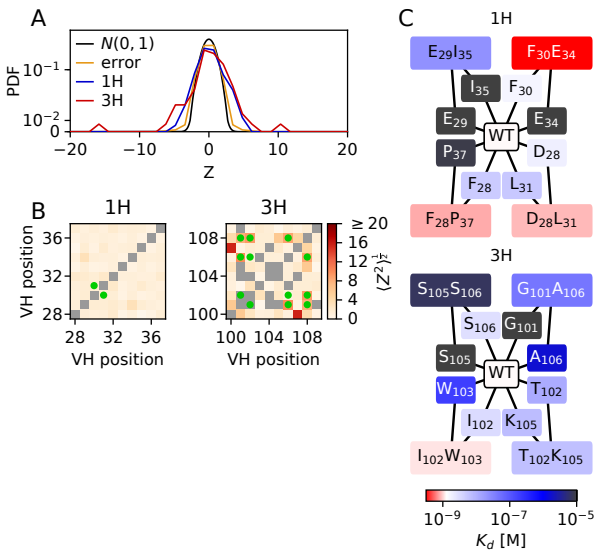

Figure S10: **The optimized PWM yields similar epistatic results to energy based PWMs, related to Figure 2.** We used the monotonic transformation (E) in figure S6 to train the PWM and test for epistasis, in contrast to binding free energy. (**A**) Distribution of Z-scores, defined as the normalized deviation from the PWM prediction, $Z_{\text{epi}} = (E_{\text{PWM}} - E)/\sqrt{\sigma^2 + \sigma^2_{\text{PWM}}}$, where σ^2 and σ^2_{PWM} are the estimated errors on E and E_{PWM}, and K_d at boundaries are removed. Positive Z-scores indicate epistasis increased affinity. The Z score standard deviation was much higher than expected from measurement errors (Z_{error}) for CDR1H (1.51, $p < 10^{-7}$, Levene's test) and CDR3H (3.11, $p < 10^{-37}$), meaning that the discrepancy between the PWM and measurement is mainly due to true epistasis. The measurement error was nearly unchanged, with standard deviation of 1.11. (**B**) Standard Z-score deviation for each pair of positions along the sequence. This deviation is higher at pairs of positions mutated in the super-optimized 4m5.3 antibody (green dots) in 3H ($p = 0.006$, Mann-Whitney), but not in 1H ($p = 0.19$). (**C**) Representative examples of sign epistasis as identified by Z-score. 51 examples of sign epistasis were found based on optimized PWMs. See Fig. 2B,C for comparison.

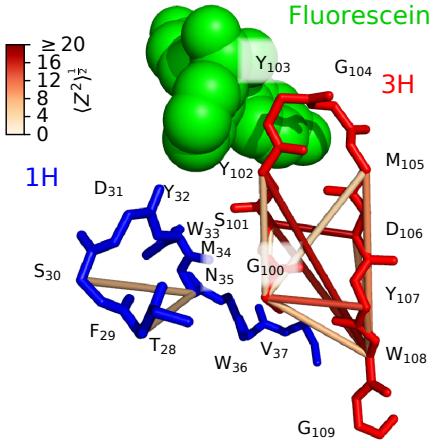

Figure S11: **Structural representation of epistatic hotspots, related to Figure 2.** Pairs of positions with standard epistatic effect $\langle Z^2_{\text{epi}} \rangle^{1/2} > 3$ (with the mean taken over all measured amino-acid variants at the two positions) are superimposed on the wildtype antibody structure. Links correspond to epistatic hotspots in Fig. 2C.

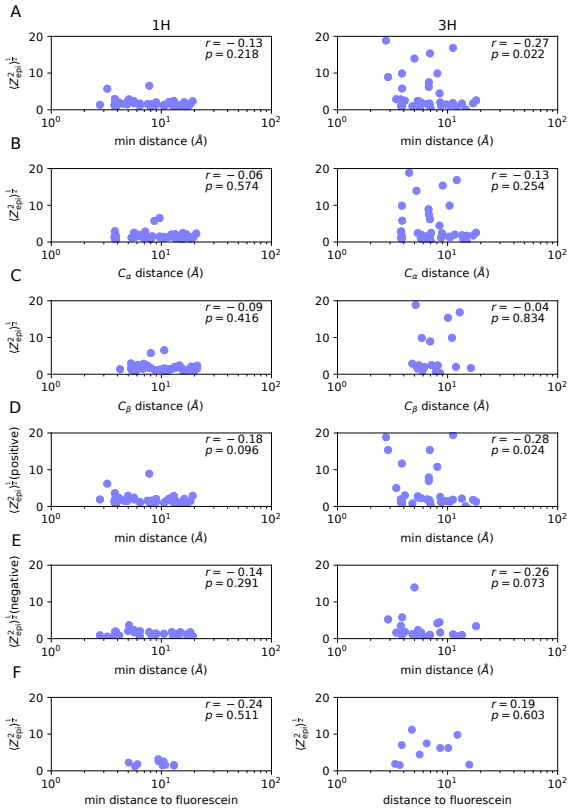

Figure S12: **Epistasis versus residue distance, related to Figure 2.** Standard epistatic effect $\langle Z_{\mathrm{epi}}^2 \rangle^{1/2}$ versus distance between residues in the 1H and 3H domains. bf A) Distances are calculated as minimum distance between any C,N, or O atom in two residues, B) between alpha carbons, and between C) beta carbons. We next compared only D)positive and E) negative epistasis against any two nearest C,N, or O atoms between two residues. F) Minimum distance between fluorescein and residue had no statistically significant correlation to average epistatic effect. See primary structure epistatic effects in Fig. 2C for comparison.

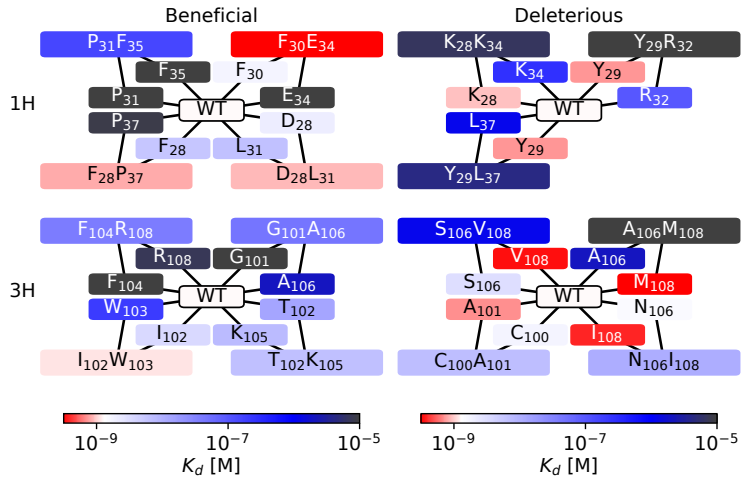

Figure S13: **Examples of sign epistasis, related to Figure 2.** Beneficial (left) and deleterious (right) sign epistasis examples are shown for the 1H and 3H domains. All 52 examples of beneficial sign epistasis with double mutant $K_d \leq 10^{-6}$ may be found in `S1_table_sign_epistasis.csv`, and summarized in tables S1 and S2. Sign epistasis represents a more extreme case of epistasis as shown in Fig. 2A-C.

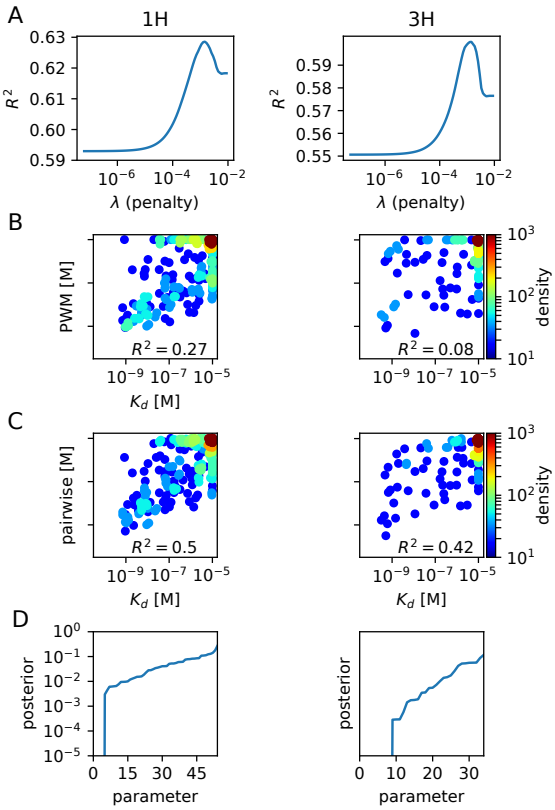

Figure S14: **Inferrence of the epistatic model, related to Figure 2.** Left panels correspond to the 1H domain, and right panels to the 3H domain. Parameters were fit by minimizing the mean squared model error with a L1 penalty on the parameters with coefficient λ in Fig. **??**D,E. (**A**) The cross-validated coefficient of determination ($1 -$ standard error^2/standard deviation^2) has a clear maximum as a function of λ. (**B**) and (**C**) Model prediction versus measurement of $F = \ln(K_d/c_0)$, for sequences involving a non-zero interacting term, when using (**B**) the PWM model and (**C**) the epistatic model with optimal λ. (**D**) Rank-ordered posterior probability of $\tilde{J}$ parameters to be non-zero, according to the Bayesian Lasso posterior (Park and Casella, 2008). Monte Carlo Markov Chains (MCMC) were used to estimate posterior probability of a parameter switching signs. Multiple ($2 \times$ number of non-zero parameters from Lasso optimization + 2) MCMCs were initialized ("thermalized") for 227 steps $\times$ the number of parameters. After thermalization, the MCMCs were simulated for another 454 steps $\times$ the number of parameters. Parameter values were sampled at intervals where their autocorrelations were ≤ 0.1. The posterior probabilty was calculated as the fraction of time the parameter switched signs in the MCMCs.

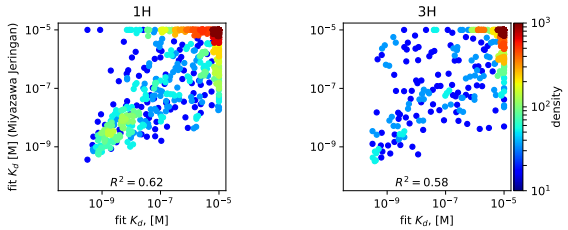

Figure S15: **Miyazawa-Jernigen predictions, related to Figure 2.** Miyazawa Jernigen contributions were scaled and translated to maximize their predictions of epistatic interactions. Residues were classified as contacting if their average atom position was less than 6.5 Å, excluding adjacent residues. These contributions explain less of the variation than from the coarse grained model in Fig. **??**D,E.

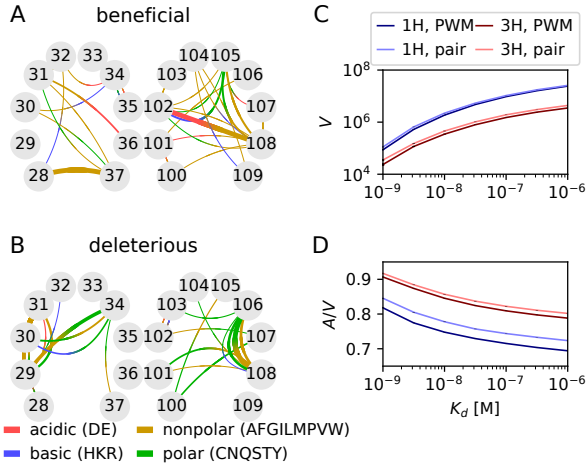

Figure S16: **Epistatic model on optimally transformed data, related to Figure 2.** A model of biochemical epistatic interactions between polar, nonpolar, acidic, and basic residues was fitted to the monotonically transformed data (Fig S6) using LASSO regularization and tested by cross-validation, yielding 1058 CDR1H and 1066 CDR3H interaction terms. Mean (**A**) beneficial and (**B**) deleterious interactions calculated by averaging over all double mutants, colored by interaction type. Line width denotes interaction strength. (**C**) Number V of amino-acid sequences of the 1H (blue) and 3H (red) regions with dissociation constant below K_d, as estimated by the PWM model (dark color) or the epistatic model (light color). Epistasis enlarges the number of variants with good affinity for both 1H and 3H. (**D**) Mutational flux A (defined as the average number of random mutation events from all possible sequences to cause the dissociation constant to cross K_d), normalized by V, showing that epistasis also increases the accessibility of the region of good binders in sequence space. Differences between the PWM and epistatic models were robust to errors in the estimate of the interaction parameters ($p < 10^{-5}$, Jackknife analysis).The optimized transformation results in small differences to the energy based PWM results shown Fig. **??**D-G